# Data bibliography

## Atmosphere / World

**01 Hajime Narukawa — AuthaGraph formulation**
2022 formulation; native Cartofreako implementation
Projection and graticules are mathematical constructions. Horizontal periodicity supports the ST2 framing study.
https://doi.org/10.11212/jjca.60.1_1

**02 Natural Earth — 1:10m physical vectors**
Pinned local vectors used by the 29 October 2026 delivery; not new observations
Land, coastline, minor islands, lakes, rivers, ice and generalized bathymetry. Individual version files and component hashes are in the JSON ledger.
https://www.naturalearthdata.com/downloads/10m-physical-vectors/
Attribution / use: Public domain

**03 JAXA P-Tree Himawari-9 Level-2 Cloud Property 1.0**
29 October 2026, 18:20 UTC
Himawari full disk; 10-minute, approximately 5 km, daytime retrieval only
ftps://ftp.ptree.jaxa.jp:990/pub/himawari/L2/CLP/010/
Attribution / use: Provider attribution and reuse conditions: https://www.eorc.jaxa.jp/ptree/terms.html

**04 GCOM-C/SGLI daily daytime aerosol optical depth at 500 nm**
15 October 2026, 00:00 UTC
global daily observed aerosol column loading; not smoke attribution or PM2.5 exposure
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

**05 JAXA GSMaP daily gauge-adjusted precipitation rate**
26 October 2026, 23:00 UTC
60S to 60N daily precipitation-rate field in mm h-1; not flood or extreme-event counts
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

**06 JASMES Aqua/MODIS daily surface shortwave radiation**
15 October 2026, 23:59 UTC
global daily surface shortwave radiation with publication lag
https://data.earth.jaxa.jp/stac/cog/v1/catalog.json
Attribution / use: Provider attribution and reuse conditions: https://earth.jaxa.jp/en/data/policy/

**Measure and interpretation**
Physical observations / 15–29 October 2026
H3 resolution 5; per-layer quality masks and valid-sample threshold retained. Source times differ. Solar geometry is calculated at 29 October 2026, 20:00 UTC. Land fill opacity zero, outlines above data.

**Source ledger**
Pinned input files, complete source manifests and checksums:
https://bdekoz.github.io/cartofreako/docs/pages/gallery/setouchi-2028-bibliography.json

H3r5 resolution trial · original observation dates and sampling retained

Actual H3 cell boundaries; original source-center values reaggregated directly at this resolution.

Unoccupied cells have no sampled center: unobserved, not zero. Source-pixel footprints are not filled.

Finer indexing adds no observations. Data layers render at 300 dpi; PDF embeds them. Outlines and type remain vector.

Atmosphere: maximum 720 samples per raster axis, quality masks retained; radiation uses the same 100 opacity levels.

## Physical observations / 15–29 October 2026

Land outlines above data; no land fill. Missing observations are not zero.

*JAXA/EORC observations; NASA MODIS shortwave input.*

Shortwave radiation · 0–400 W/m² · 100 opacity bands · 15 Oct

Aerosol optical depth · 500 nm · 15 Oct

Precipitation · mm/hour · 26 Oct

Confident-cloud fraction · 29 Oct

Cloud optical thickness · 29 Oct

Cloud-top height · km · 29 Oct

ISCCP cloud type · 29 Oct

Cloud retrievals are regional and daytime only. AOD is not smoke or PM2.5.

Antarctica and its terrestrial outlines, ice and country fills are deliberately displaced to the lower left. This is an editorial position; geophysical fields and network observations retain their source coordinates.

AuthaGraph frame shifted horizontally by 10% of image width. All artwork scaled to a 43.5-inch image on the 44-inch sheet.

Radiation opacity · 100 bands

Stronger cells are more opaque; the same scale applies to both views.

0 W/m² · 0%  200 W/m² · 50%  400 W/m² · 100%

Low values fade to transparent. Missing observations are not zero.

Solar / cloud Atmosphere